Customer spending forecasts

90-day and 180-day purchase value for marketing prioritization

This project predicts recorded future purchase value from prior transaction history and evaluates customer selection under a uniform contact budget. It uses a consented retrospective cohort and a local, reproducible Python pipeline.

What the untouched tests show

Scoring date	Days	RMSE	WAPE	Top 20% capture
2022-01-01	90	$528	58.9%	53.4%
2022-05-01	90	$553	51.4%	53.6%
2022-01-01	180	$887	55.0%	53.2%
2022-05-01	180	$870	46.3%	52.9%

Validation selected histogram gradient boosting for 90 days and Ridge with alpha 100 for 180 days. Both final models use labels available before January 2022 and remain fixed for the January and May tests.

The selected models improve RMSE against the trailing-year baseline on all four test evaluations. Their top-20% capture gains over raw historical-spending ranking are modest: **0.28 to 0.99 percentage points**. The January 180-day model overpredicts aggregate spending by **25.3%**.

A longer horizon has better normalized errors in these tests, while the 90-day model has stronger aggregate calibration. Horizon choice should follow the decision cadence and these tradeoffs. Smaller short-horizon dollar error alone is insufficient.

Decision scope: customer-value coverage under a contact budget. Campaign uplift, incremental revenue, profit CLV, and causal marketing ROI require additional data.

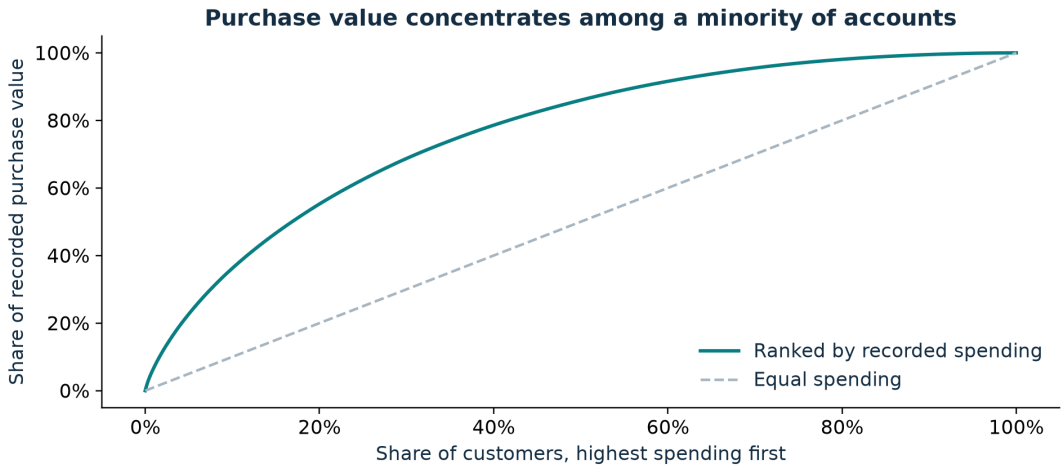
Data provenance and quality

Open e-commerce 1.0 contains crowdsourced Amazon purchase histories from consenting U.S. participants. Harvard Dataverse publishes version 1.0 under CC0 1.0. The mirror's bytes match the original release's MD5.

Audit item	Measured result
Raw file	1,850,717 rows / 5,027 accounts
Retained common window	1,737,058 rows / 5,015 accounts
Purchase value	$41,258,954
Later dates excluded	113,659 rows
Identical rows retained	10,911
Gift/reload title matches	39,006 rows

Valid transactions span January 1, 2018 through October 31, 2022. The raw file extends to August 2024, including one anomalous 2024 record. The common exclusive end of November 1, 2022 avoids treating uneven later exports as equivalent coverage.

No rows failed customer, date, price, or quantity checks in this file. Missing category, title, product, and state values become explicit UNKNOWN descriptions. Unit prices use Decimal half-up cent rounding, quantities must be positive integers, and value aggregates in integer cents. Equal rows remain because no order ID resolves duplicates.



MD5: 3549d72735b1211ab009ff0177b94bd9

Sources: Berke et al. (2024), Scientific Data 11, 491. doi:10.1038/s41597-024-03329-6. Dataset: doi:10.7910/DVN/YGLYDY. Exact metadata and SHA256 accompany this project.

Customer snapshots and targets

Monthly scoring dates begin January 2019. Eligibility requires at least one valid purchase before scoring. One account-date is one modeling observation. Accounts repeat over time, so observations and overlapping forecast windows are dependent.

Component	Contract
Historical features	Transaction date strictly before scoring date
90-day target	Sum over [scoring date, scoring date + 90 days)
180-day target	Sum over [scoring date, scoring date + 180 days)
No future purchases	Recorded zero if the entire window is observed
Incomplete future window	Unavailable target, never an assumed zero

A January 1 purchase enters the target, while December 31 remains historical. January 1, 2022's 90-day target ends April 1 exclusively. Its 180-day target ends June 30 exclusively. These are fixed day counts, not calendar-month approximations.

Purchase-derived features

Recency and observed history length describe engagement and available tenure. For 30, 90, 180, and 365 days, features measure spending, quantities, distinct purchase days, product/category diversity, and spending per purchase day. Trends compare recent and preceding spending. Month sine/cosine features encode seasonality.

A frozen vocabulary learns the top 12 categories by 2018 spending, before every snapshot. UNKNOWN handles missing categories and OTHER handles new categories. Ridge standardization fits only on each fold's training examples. Identifiers and later-collected survey answers never enter predictive inputs.

The package shares the same feature function and prediction bundle between training and batch scoring. Tests append future purchases and check that all earlier features remain identical.

Label-aware chronological validation

A historical feature row can still leak if its future label was unavailable at the validation date. The pipeline explicitly requires every training target's exclusive end to be strictly earlier than its cutoff. This conservative purge protects both horizons.

Stage / horizon	Cutoff	Latest train label end	Training rows
validation / 90d	2021-01-01	2020-12-30	101,010
validation / 90d	2021-07-01	2021-03-01	110,740
validation / 180d	2021-01-01	2020-12-28	86,523
validation / 180d	2021-07-01	2021-05-30	110,740
final fit / 90d	2022-01-01	2021-12-30	159,960
final fit / 180d	2022-01-01	2021-12-28	145,118

Validation uses January and July 2021, with development scoring dates confined to 2019-2020. Equal weighting of the two fold RMSEs selects candidates. Once those choices freeze, final fitting can use matured 2021 examples whose labels finish before January 2022.

The same fitted final bundle scores January 1 and May 1, 2022. May features legitimately incorporate purchases before May. Those observed histories can overlap January outcomes, but they never become final-model training labels.

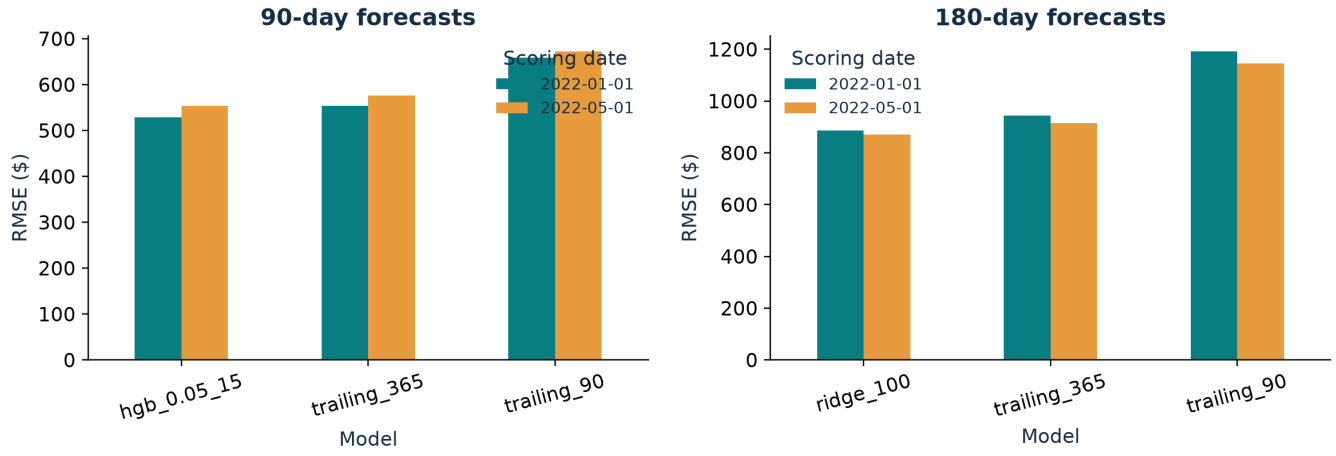
Both tests are historical demonstrations on a selected cohort. Repeated accounts and overlapping target windows prevent interpreting the two dates as independent experiments. The bootstrap preserves account clusters, with calendar-regime uncertainty still unresolved.

Models and test performance

Each horizon compares trailing-90 and trailing-365 spending rates, Ridge alpha 0.1/1/10/100, and histogram boosting at learning rate 0.05/0.1 with 15/31 leaves and 300 iterations. Random internal early stopping is disabled. Seed: 42.

Choose lowest mean validation RMSE. Within 1%, prefer top-20% capture, then simplicity. ML must improve the best baseline's RMSE to qualify. Clamp forecasts at zero. Select 90 days first, then evaluate 180-day candidates after max(raw180, selected90). Apply that same constraint at test and scoring.

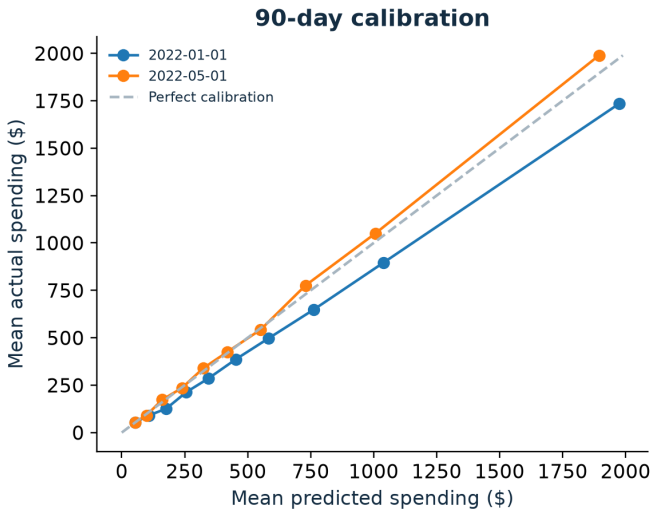
Test errors for selected models and spending baselines

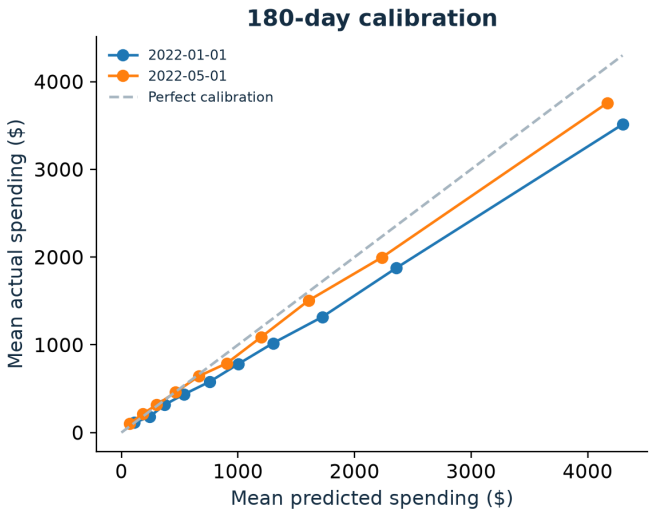


Date	Days	Selected model	RMSE vs 365d baseline
2022-01-01	90	hgb_0.05_15	4.6% lower
2022-05-01	90	hgb_0.05_15	3.8% lower
2022-01-01	180	ridge_100	6.0% lower
2022-05-01	180	ridge_100	5.0% lower

For 180-day model comparisons, each candidate receives the same horizon-coherence adjustment. Raw historical-spending ranking remains a separate benchmark. No test results feed model selection.

Calibration and spending capture





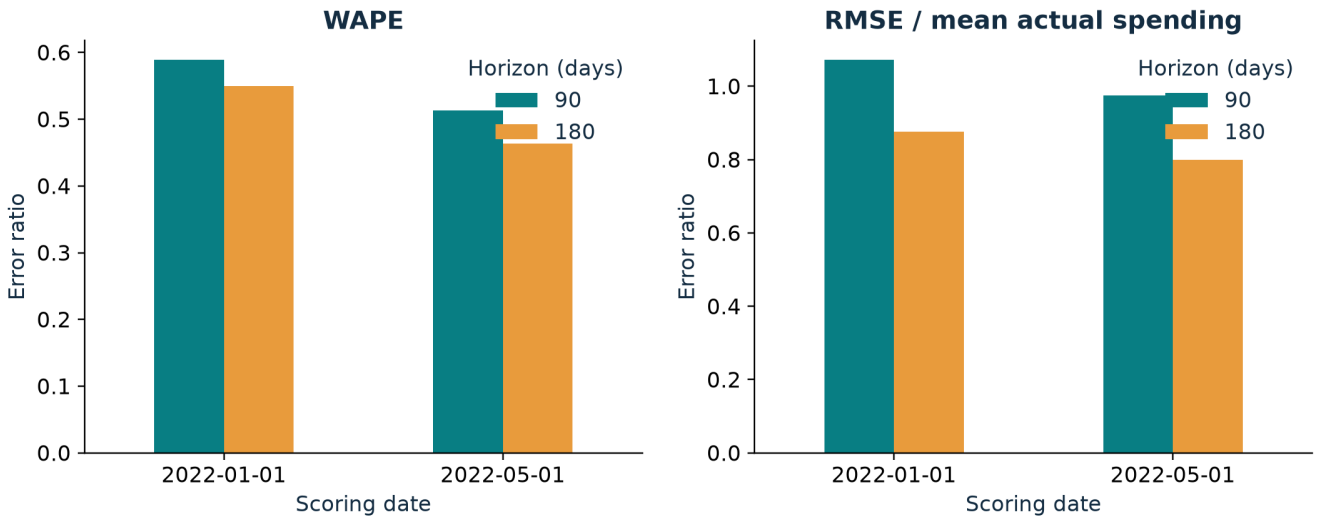
Deciles compare mean predicted and actual spending within ranked groups. A model can rank valuable accounts reasonably well while overstating their dollar value. Aggregate predicted/actual ratios expose an important shortcoming here.

Date	Days	Predicted / actual total	Top-20% gain vs history
2022-01-01	90	1.166	+0.28 pp
2022-05-01	90	0.965	+0.99 pp
2022-01-01	180	1.253	+0.99 pp
2022-05-01	180	1.085	+0.59 pp

The selected rankings capture about 53% of future spending in the top fifth of accounts. Most of that concentration already appears in historical-spending rankings. The additional gain from modeling is small and should be presented alongside uncertainty, not as a large marketing lift.

The 180-day adjustment raises 1.34% of January forecasts and 4.38% of May forecasts. It protects arithmetic consistency, but does not guarantee calibration. Predictive associations in the saved permutation table do not establish causal drivers.

Horizon comparison and uncertainty



WAPE divides absolute error by actual dollars. Normalized RMSE divides RMSE by mean actual spending. Both normalize the spending scale and favor the 180-day horizon here. The 90-day model's aggregate ratios are closer to one. A business horizon depends on whether ranking, amount calibration, or planning cadence matters most.

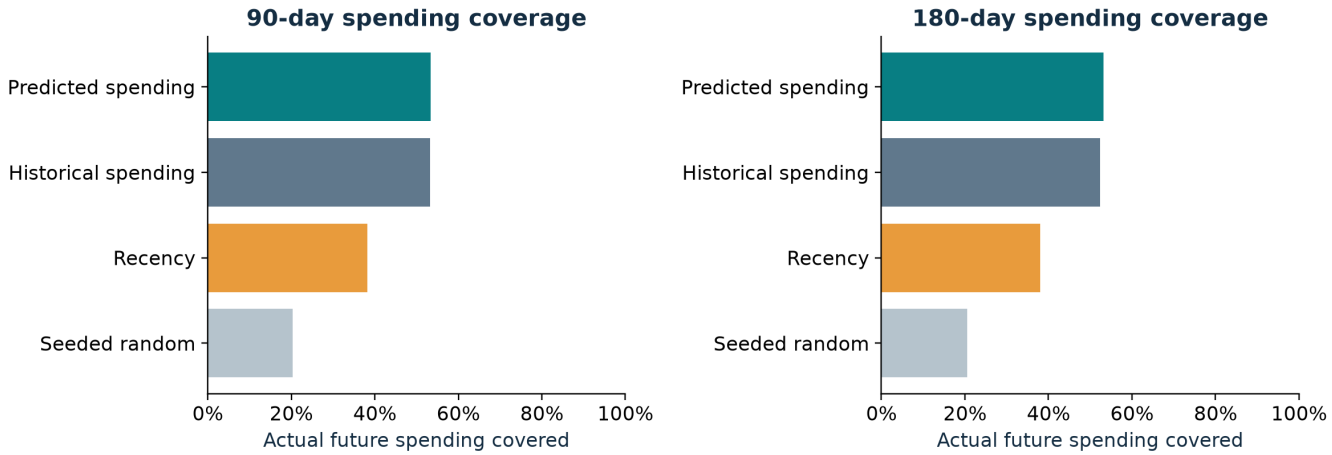
Days	Pooled metric	Estimate	95% customer-bootstrap interval
90	rmse	541.076	[499.990, 586.250]
90	wape	0.548	[0.535, 0.562]
90	capture_20	0.533	[0.521, 0.546]
90	capture_20_gain_pp	0.555	[0.030, 1.027]
180	rmse	878.467	[825.794, 928.103]
180	wape	0.505	[0.491, 0.519]
180	capture_20	0.527	[0.516, 0.540]
180	capture_20_gain_pp	0.466	[0.041, 1.020]

500 resamples draw accounts with replacement, retaining every repeated test observation for a sampled account. Percentile intervals condition on the frozen models and chosen cohort. They do not refit models or account for all model-selection, source-coverage, or future-time uncertainty. Pooled windows can count the same purchase twice and are not unique annual spending.

Marketing prioritization

Illustrative budget: $1,000. Uniform contact cost: $1. Capacity equals floor(budget / cost), capped at eligible accounts. All strategies contact the same 1,000 accounts. The simulator compares predicted spending with historical spending, recency, and seed-42 random ranking.

Illustrative $1,000 budget at $1 per contact · January 2022

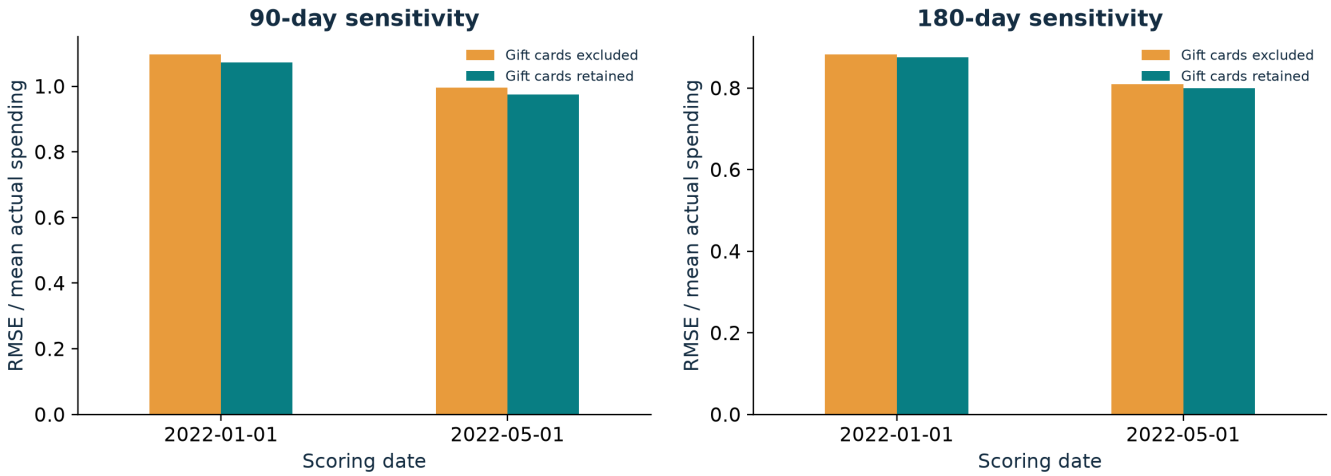


Days	Strategy	Covered future value	Coverage
90	Predicted spending	$1,313,622	53.4%
90	Historical spending	$1,309,821	53.2%
90	Recency	$941,113	38.2%
90	Seeded random	$500,477	20.3%
180	Predicted spending	$2,690,119	53.2%
180	Historical spending	$2,644,603	52.3%
180	Recency	$1,923,176	38.0%
180	Seeded random	$1,038,216	20.5%

Historical ranking overlaps 87.0% of the 90-day predicted audience. The dashboard displays and downloads the same ranked rows. Zero budgets select nobody, oversized budgets select all, and invalid budgets/costs reject. Decimal floor arithmetic protects contact capacity at cent boundaries.

Business interpretation: customer-value coverage under a contact budget. The accounts might have purchased anyway. Causal campaign return requires randomized treatment/control outcomes and margins.

Sensitivity and customer examples



The primary retains 39,006 title-matched gift/reload rows. The separately rerun sensitivity excludes the documented terms and can change targets, cohort eligibility, and model selection. Normalized RMSE becomes slightly worse in all four sensitivity evaluations, despite smaller dollar RMSE. Amount scales matter.

Date	Days	No-gift selected model	Normalized RMSE
2022-01-01	90	hgb_0.05_31	1.096
2022-05-01	90	hgb_0.05_31	0.996
2022-01-01	180	ridge_10	0.882
2022-05-01	180	ridge_10	0.809

January 2022 account examples

Forecast percentile example	Days	Predicted	Actual
Low forecast	90	$26	$27
Median forecast	90	$397	$114
High forecast	90	$1,678	$1,559
Low forecast	180	$29	$539
Median forecast	180	$882	$2,122
High forecast	180	$3,839	$1,941

Examples use forecast ranks, not favorable outcomes. January 90-day group RMSE is $211 for 35 short-history accounts, $112 for 317 inactive accounts, and $1,207 for 500 high historical spenders. Their WAPE is 103.8%, 183.9%, and 46.6%, respectively. Small dollar errors can hide large relative errors in low-spending groups. Groups overlap and small groups are unstable.

Limitations and reproducibility

What these results can support

Recorded fixed-horizon spending forecasts and retrospective ranking comparisons for established accounts in this cohort. They support discussing which customers cover more observed future value within an assumed contact budget.

What remains unresolved

The sample is consent-selected and favors established accounts still active at recruitment. Accounts can be shared. History starts in 2018 and does not reveal true tenure. Common October 2022 coverage follows the source study but transaction absence alone cannot prove complete per-account exports. The data omit usable campaign assignments, margins, true order IDs, and resolved return adjustments. Behavior in later years can differ.

Customer spending is not Amazon net revenue or profit. Predictive importance measures association. Neither spending coverage nor ranking gain establishes response to marketing, incremental revenue, causal ROI, or an optimal offer.

Reproduce and inspect

Create Python 3.12 environment, install requirements.lock.txt, then install the package. Run python -m amazon_clv run with the verified CSV and a new output directory. Run python -m amazon_clv score with a historical date and saved bundle. The README supplies exact commands. Four clean-kernel notebooks explain data quality, features, modeling, and marketing. Tests protect temporal boundaries, serialization, and budget exports.

The release includes source hashes, exclusion and split audits, frozen selections, model bundles, predictions, metric tables, calibration, bootstrap intervals, gift sensitivity, charts, a local Streamlit dashboard, data dictionary, model card, chronological work log, and interview guide.

References

Berke et al. (2024). Open e-commerce 1.0. Scientific Data 11, 491. nature.com/articles/s41597-024-03329-6
Harvard Dataverse v1.0, CC0 1.0. doi.org/10.7910/DVN/YGLYDY
Streamlit caching documentation. docs.streamlit.io/develop/concepts/architecture/caching
All figures and measured findings: this project's saved original-data run, seed 42.